

CH16: Chromatin module summary of genes

Functional group	Genes	Principal chromatin role
DNA methylation establishment and maintenance	DNMT3A, DNMT1, UHRF1	Establishing, copying, and preserving CpG methylation patterns through replication and differentiation
H3K9 methyltransferases	EHMT1, EHMT2, SETDB1, SUV39H1, SUV39H2	Writing repressive H3K9 methylation marks that support facultative or constitutive heterochromatin
H3K9me/HP1 readers	CBX1, CBX3, CBX5	Reading H3K9 methylation and promoting chromatin compaction, heterochromatin organization, and stable silencing
Silencing and repeat defense	TRIM28, SETDB1, MORC3	Repression of endogenous retroviruses, transposable elements, and other repetitive DNA
Remodeling and chromatin deposition	ATRX	H3.3 deposition and maintenance of telomeric/pericentromeric repetitive chromatin
Nuclear architecture and genome integrity	LMNA, BANF1	Nuclear-lamina organization, positioning of peripheral heterochromatin, chromosome/nuclear-envelope integrity, and DNA-damage resilience